

Personal Profile

Bioinformatician with two master's degrees, most recently building automated genome-scale metabolic modelling pipelines across roughly 1,500 bacterial genomes. Comfortable owning a pipeline end to end — reconstruction, QC, validation against experimental data, and the reproducibility discipline behind trustworthy results. Python and R daily, Linux and Git throughout.

Education

2024 – 2026	Master of Bioinformatics <i>KU Leuven — Cum Laude</i>
2022 – 2024	Master of Biomedical Sciences <i>KU Leuven — Cum Laude</i>
2019 – 2022	Bachelor Biomedische Wetenschappen <i>KU Leuven — Magna Cum Laude</i>
2017 – 2019	Bachelor Burgerlijk Ingenieur <i>KU Leuven</i>

Research

2025 – 2026	MSc Thesis — Pan-Metabolic Modelling Software for Therapeutics <i>KU Leuven · github.com/robbedewin/PaRMMoSaHN</i> - Open-source Python package: one pan-metabolic model per species, with every strain model derived from it by homology projection — minutes per genome instead of hours. - Benchmarked against the established per-genome tools on experimental substrate data: matched or beat both on accuracy at roughly 12× the throughput.
2023 – 2024	MSc Thesis — Genetic Changes and Allele-Specific Expression in Induction Failure T-ALL <i>KU Leuven</i> Snakemake WGS and RNA-seq pipelines on the T2T-CHM13 reference — variant calling, copy number and allele-specific expression, recovering the known <i>NOTCH1</i> and <i>CDKN2A/B</i> lesions plus <i>FAT1</i>.

Skills & Expertise

Technical Expertise

Languages: Python, R, Bash; LaTeX

Bioinformatics: Genome-scale metabolic modelling (gapseq, CarveMe, memote), flux balance analysis, pangenome analysis (PPanGGOLiN, DIAMOND), WGS/RNA-seq variant calling (Mutect2, GRIDSS, ASCAT)

Data & ML: Machine learning and neural networks, Bayesian modelling, high-dimensional data analysis, statistical validation

Engineering: Snakemake, Git/GitHub, Linux, conda, pytest, Quarto, reproducible pipeline design, long-running batch workloads

Software Projects

Self-built financial planning toolchain: a Quarto report generator, an audit tool with alerting, a pytest suite and an ISIN-keyed instrument registry. github.com/robbedewin

Work Experience

2017 – 2025	Student summer job — Belcar (automotive)
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Robbe De Win

Personal Details

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LinkedIn
linkedin.com/in/robbedewin

GitHub
github.com/robbedewin

Driving Licence: B

Languages

Dutch

Native

English

Full professional proficiency

French

Basic

Software

Python

Daily driver

R

Statistics and visualisation

Linux & Git

Everyday working environment

Interest

Cycling

Football

Skiing